Chapter 30

Convexity, Timing, and Quanto Adjustments †

Almost everything in the preceding chapters rests on the same approach: when the expected future value of a rate or a price is required, today's forward value is used in its place. Forward contracts were valued that way, swaps were valued that way, and Chapter 29 priced caps and swaptions by assuming the relevant forward rate was the mean of a lognormal distribution.

That approach is valid far more widely than one might expect, but it is not universal. It is the consequence of a specific and rather delicate alignment: the forward value of a variable equals its expected future value *only under one particular probability measure*, and which measure that is depends on when the payoff is made and in what currency.

When a contract respects that alignment (a rate observed at the start of a period and paid at the end of it, in its own currency), the forward rate is exactly the right expectation and no adjustment is needed. When a contract breaks the alignment, the forward rate is the wrong expectation, and using it introduces a pricing error that can run from a fraction of a basis point to tens of basis points.

This chapter is about the three ways the alignment breaks and the corrections each one requires:

- a convexity adjustment, when the contract pays a rate linearly although the instruments that define its forward are not linear in it: a futures rate under daily margining, a rate paid at the start of its period, a swap rate paid on a single date;
- a timing adjustment, when a payoff is made at a date other than the one naturally associated with the variable;
- a quanto adjustment, when a payoff denominated in one currency is based on a variable belonging to another.

All three are the same phenomenon in a different form, and Section 30.1 explains why.

These adjustments are not academic refinements. The convexity adjustment on a three-month rate futures contract expiring in ten years can reach 40 to 50 basis points—larger than the bid–ask spread by two orders of magnitude, and large enough that a curve built without it will misprice every long-dated swap on the book.

The chapter runs from the principle to the three corrections. Section 30.1 states when a forward value is an expectation and under which measure. Sections 30.2 to 30.5 take the breaks in turn: a payment made late, the daily settlement that separates a futures contract from a forward, a rate paid at the start of its period, and the swap rate paid outside its own annuity, which is the largest of the adjustments that come from payment timing. Section 30.6 is the currency break, and Section 30.7 shows the three as one calculation done three times. Sections 30.2 through 30.6 can be read in any order, and each adjustment is derived where it is stated; those derivations rest on Chapter 28 and can be skipped without losing the results.

30.1 FORWARD VALUES ARE EXPECTATIONS—UNDER THE RIGHT MEASURE

The general principle comes first, because the three adjustments that follow are special cases of it. The chapter requires a single concept, Chapter 28's numeraire: any asset with a strictly positive price in which all other prices are measured. What follows is the working statement of it, and nothing here requires more. If $N(t)$ is the price of a traded asset that pays no income, then there exists a probability measure (call it Q^N, the measure *associated with* the numeraire N) under which the price of every other traded asset, divided by N, is a martingale: for any traded asset with price $V(t)$,

$$\frac{V(t)}{N(t)} = E^N\left[\frac{V(T)}{N(T)}\,\middle|\,\mathcal{F}_t\right], \qquad t \le T$$

where E^N denotes expectation under Q^N and $\mathcal{F}_t$ the information available at t. This is the fundamental result of arbitrage pricing, and everything in this chapter follows from it. Only the case $t = 0$ is used below; the martingale property is the stronger statement that the equality holds at every intermediate date as well, which is what makes a change of numeraire legitimate at any point in a contract's life. Two choices of numeraire dominate practice.

The money market account. Take $N(t) = B(t)$, the value of a bank account earning the instantaneous short rate. The associated measure is the familiar risk-neutral measure Q, in which expected returns equal the risk-free rate.

The zero-coupon bond. Take $N(t) = P(t,T)$, the price at time t of a bond paying \$1 at time T. The associated measure is called the **T-forward measure**, written Q^T. It has a property that makes it enormously convenient: since $P(T,T) = 1$,

$$V(0) = P(0,T)\,E^T[V(T)] \tag{30.1}$$

The value today of a payoff received at time T is the discount factor times the expected payoff under the T-forward measure: no stochastic discount factor appears inside the expectation, and discounting and expectation separate cleanly. This is why the T-forward measure is the natural home for any contract with a single payment date T.

Which rates are martingales, and under which measure

Now consider the forward rate. Let $F(t;T_1,T_2)$ be the forward rate observed at time t for the period $[T_1,T_2]$, with accrual factor $\tau = T_2 - T_1$, defined in the usual way from bond prices:

$$F(t;T_1,T_2) = \frac{1}{\tau}\left[\frac{P(t,T_1)}{P(t,T_2)} - 1\right] = \frac{P(t,T_1) - P(t,T_2)}{\tau\,P(t,T_2)} \tag{30.2}$$

Look at the right-hand expression. The numerator, $P(t,T_1) - P(t,T_2)$, is the price of a traded portfolio: long one T_1-bond, short one T_2-bond. The denominator is τ times the price of the T_2-bond. So F is the price of a traded asset divided by τ times a numeraire, and therefore F is a **martingale under Q^{T_2}**:

$$E^{T_2}[F(T_1;T_1,T_2)] = F(0;T_1,T_2) \tag{30.3}$$

At $t = T_1$ the period begins immediately, so $F(T_1;T_1,T_2)$ is the spot rate for $[T_1,T_2]$, which later sections write $R(T_1)$. Equation (30.3) is therefore the precise statement of the approach relied on throughout: the forward rate is the expected future spot rate, but only under the T_2-forward measure, where T_2 is the date the rate is *naturally* paid.

The two results together give the chapter its subject. Equation (30.1) requires an expectation under Q^T to value a payoff received at time T; equation (30.3) supplies an unbiased expectation under Q^{T_2}. If the payment date T and the natural date T_2 coincide, the two align and the

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forward rate is used directly. If they do not, because the payment comes early, or late, or in a different currency, or because margining has changed the effective numeraire to the money market account, then the required expectation is under a different measure (Q^T if the payment moves, Q if margining makes the money market account the numeraire), while only $E^{T_2}[F]$ is known, and the difference is the adjustment. Call the measure under which the variable's forward value is an unbiased expectation its **natural measure** (here Q^{T_2}) and the measure the contract's payment calls for the **required measure**; each section below names both.

Why an adjustment is a covariance

Changing measure is done with a Radon–Nikodym derivative, and for forward measures it takes a particularly interpretable form. Moving from Q^{T_2} to Q^T requires reweighting each scenario by the ratio of the two numeraires' growth:

$$\left.\frac{dQ^T}{dQ^{T_2}}\right|_{\mathcal{F}_t} = \frac{P(t,T)\,/\,P(0,T)}{P(t,T_2)\,/\,P(0,T_2)} \tag{30.4}$$

Write Λ_t for this ratio (Chapter 28's density process, with Q^{T_2} in place of P and Q^T in place of Q) and Λ for its value at the date the variable is observed. It is not constant: the two time-zero prices are fixed, so it moves with $P(t,T)/P(t,T_2)$, which is the discount factor between the two dates, not with either bond price on its own, since a rise in rates lowers both. When the required date is the later one, $T > T_2$, that factor falls as rates rise, so high-rate scenarios are downweighted and low-rate scenarios upweighted; when $T < T_2$ the weighting runs the other way. The reweighting is monotone in rates either way, and that is what matters here. Therefore:

> **If the variable being adjusted is correlated with interest rates, reweighting the scenarios changes its mean. If it is uncorrelated, the mean is unchanged and no adjustment is needed.**

Every adjustment in this chapter is a covariance with that density, and at leading order a covariance between the variable and whatever drives it: an interest rate, or an exchange rate. The sign of that covariance gives the sign of the adjustment, which is enough to check any formula in this chapter without re-deriving it.

Section 28.6 derives (30.4) and one consequence. Because the density has unit mean, for any X known at t,

$$E^T[X] = E^{T_2}[X] + \mathrm{Cov}^{T_2}(X, \Lambda_t) \tag{30.5}$$

Every derivation in this chapter is an evaluation of (30.5) for a particular X, and the work each time is finding what Λ_t is a function of.

30.2 TIMING ADJUSTMENTS

Start with the simplest break in the alignment: a payoff based on a variable observed at time T^*, but paid at some later time T rather than at T^*. Let V be the variable, with known forward value V_0, meaning $E^{T^*}[V(T^*)] = V_0$. What is required is $E^T[V(T^*)]$.

Write $D = P(T^*,T)$ for the discount factor over the deferral period and $D_0 = P(0,T)/P(0,T^*)$ for its forward value today. Evaluating (30.4) at $t = T^*$ and using $P(T^*,T^*) = 1$, the density is simply the discount factor rescaled,

$$\Lambda \;=\; \frac{P(T^*,T)\,/\,P(0,T)}{P(T^*,T^*)\,/\,P(0,T^*)} \;=\; \frac{P(T^*,T)\,P(0,T^*)}{P(0,T)} \;=\; \frac{D}{D_0},$$

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and $E^{T^*}[D] = D_0$ because $P(t,T)/P(t,T^*)$ is a traded asset over the numeraire. So (30.5) reads

$$E^T[V(T^*)] \;=\; V_0 \;+\; \frac{\mathrm{Cov}^{T^*}(V,D)}{D_0} \tag{30.6}$$

exactly. The **timing adjustment** is the covariance of the payoff with the discount factor that defers it, and nothing has been approximated yet.

What has to be approximated is the translation into quoted terms, because the market quotes the volatility of a rate rather than of a discount factor. Writing $\tau = T - T^*$ and y for the simply-compounded rate covering $[T^*, T]$, the two bond prices are defined by that rate,

$$D = \frac{1}{1+y\tau}, \qquad D_0 = \frac{1}{1+y_0\tau},$$

so $\Lambda = (1+y_0\tau)/(1+y\tau)$, with $\Lambda(y_0) = 1$ and $\Lambda'(y_0) = -\tau/(1+y_0\tau)$. Linearizing Λ about y_0 turns (30.6) into a covariance with the rate,

$$E^T[V] \;\approx\; V_0 - \frac{\tau}{1+y_0\tau}\,\mathrm{Cov}^{T^*}(V,y),$$

and for jointly lognormal V and y that covariance is approximately $V_0 y_0 \rho \sigma_V \sigma_y T^*$, giving:

$$E^T[V(T^*)] \;\approx\; V_0\left[1 - \frac{\rho\,\sigma_V\,\sigma_y\,y_0\,\tau\,T^*}{1+y_0\tau}\right] \tag{30.7}$$

or, in the exponential form the market usually quotes, which agrees with (30.7) to first order and is exact when V and D are jointly lognormal ([TN30] shows why, and how much the linearization costs),

$$E^T[V(T^*)] \;\approx\; V_0 \exp\!\left(-\frac{\rho\,\sigma_V\,\sigma_y\,y_0\,\tau\,T^*}{1+y_0\tau}\right) \tag{30.8}$$

where

- y_0 is today's forward rate for the deferral period $[T^*, T]$, simply compounded over τ;
- σ_y is the volatility of that forward rate (proportional, that is, lognormal);
- σ_V is the volatility of the forward value of V;
- ρ is the correlation between V and y.

Reading the formula. The adjustment is negative when $\rho > 0$. The intuition is exactly the covariance argument of Section 30.1: deferring a payment means discounting it over $[T^*, T]$ at a rate that is not known today. If V tends to be large precisely when that rate is high, then the large payoffs are the ones that get discounted most heavily, and the deferred claim is worth less than a naive calculation suggests. If V and rates are uncorrelated, $\rho = 0$ and no adjustment is needed no matter how long the deferral.

Note also that the adjustment scales with τT^*, the product of the *deferral length* and the *time until observation*. A long deferral of a payoff observed tomorrow needs almost no adjustment, because there is almost no time for V and y to move together.

Example 30.1 Payment deferred by one quarter

A structured note pays the five-year swap rate observed at the end of year 5. Payment, however, is made at the end of year 5.25 rather than immediately on observation.

The swap rate's expected value for payment at year 5 (its forward plus the CMS adjustment of Section 30.5, taken as a round $V_0 = 4.00\%$) has volatility $\sigma_V = 20\%$. The forward rate covering the deferral quarter is $y_0 = 4.00\%$ with volatility $\sigma_y = 20\%$. The correlation between the two is $\rho = 0.7$. Here $T^* = 5$ and $\tau = 0.25$.

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The exponent in (30.8) is

$$-\frac{0.7 \times 0.20 \times 0.20 \times 0.04 \times 0.25 \times 5}{1 + 0.04 \times 0.25} = -0.001386$$

so

$$E^T[V] = 4.00\% \times e^{-0.001386} = 3.9945\%$$

The timing adjustment is −0.55 basis points.

The half basis point of Example 30.1 may look like a rounding error, and on a single trade it is. On a $5 billion structured note program it is $275,000 per year of exposure, and it is the kind of error that is systematically in one direction, so it never averages out across the book.

30.3 CONVEXITY ADJUSTMENTS I: FUTURES VERSUS FORWARDS

The largest and most economically important convexity adjustment in fixed income arises from a difference in settlement mechanics rather than payment dates. It is the reason a futures rate is not a forward rate.

Why margining changes the measure

Recall Chapter 2. A forward contract settles once, at maturity. A futures contract is marked to market daily: gains and losses are paid in cash as they accrue.

That difference determines the numeraire. Because the cash flows on a futures contract occur continuously and are immediately available for investment (or must be immediately funded), the correct numeraire is the **money market account**, and Section 28.7 showed that the futures price is then a martingale under the risk-neutral measure Q. The forward price is a martingale under Q^T. So for an asset settling at T,

$$\text{Futures price} = E^Q[S_T], \qquad \text{Forward price} = E^T[S_T] \tag{30.9}$$

Two measures, two expectations of the same terminal price, and (30.5) compares them. The density taking Q to Q^T is the T-bond over the money market account, normalized,

$$\Lambda \;=\; \frac{P(T,T)\,/\,P(0,T)}{B(T)\,/\,B(0)} \;=\; \frac{1}{B(T)\,P(0,T)}$$

so

$$\text{Forward price} - \text{Futures price} \;=\; \text{Cov}^Q(S_T,\,\Lambda) \tag{30.10}$$

and Λ is large exactly when $B(T)$ is small, that is, when rates have been low. An asset positively correlated with interest rates therefore has a futures price above its forward price, and one negatively correlated has a futures price below it. The sign follows from the covariance alone; no model of rates has been assumed.

Which is larger, and why

Section 5.12 settled the direction from the margin flows alone, before any of the measure language was available:

$$\text{Futures price} \;>\; \text{Forward price} \qquad \text{(asset positively correlated with rates)}$$

with a negative correlation reversing it. Equation (30.10) verifies that argument: the covariance it exhibits carries exactly the sign Chapter 5 reasoned to from the cash flows, and, being an expectation, it can also be evaluated.

An interest rate futures contract is the extreme case, which is why the adjustment is largest here: its underlying is the rate itself, so the correlation is perfect rather than incidental. The

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futures rate, the rate the quoted price implies (100 less the quotation, for the contracts of Section 6.2), is an affine function (a constant plus a fixed multiple) of the futures price and so is itself a martingale, so (30.9) becomes

$$\text{Futures rate} = E^{Q}[R(T_1)], \qquad \text{Forward rate} = E^{T_2}[R(T_1)] \tag{30.11}$$

the second by (30.3). Since R is the rate itself, the inequality above applies unchanged: the futures rate exceeds the forward rate by the convexity adjustment.

The size of the adjustment

Under the Ho–Lee model (the simplest no-arbitrage short-rate model, developed with its mean-reverting extension in Sections 31.5 and 31.6), the short rate follows

$$dr = \theta(t)\,dt + \sigma\,dW$$

with $\theta(t)$ a drift chosen so that the model reproduces today's term structure and σ the standard deviation of the *absolute* (not proportional) change in the short rate per year, and the adjustment takes a memorable form:

$$\text{Forward rate} \;=\; \text{Futures rate} \;-\; \frac{1}{2}\,\sigma^2\,T_1\,T_2 \tag{30.12}$$

where T_1 is the time until the futures contract's rate is determined and T_2 is the time until the end of the period the rate covers.

The structure of (30.12) carries the lesson. The adjustment grows with the square of maturity, so it is negligible for near-dated contracts and dominant for far-dated ones. Chapter 6 stated the same formula and worked it at a normal volatility of 100 basis points a year (Section 6.2 and Example 6.12): 0.63 basis points on the one-year contract, 33 at eight years and about 51 at ten. Strictly, (30.12) and (30.13) adjust the continuously compounded rate; for the simply compounded quarterly rate on which the contract settles, add about $\frac{1}{2}\sigma^2\tau T_1$, which is 0.1 basis points at one year and under 3% of the adjustment at ten.

Ho–Lee overstates the long end, because it has no mean reversion and mean reversion damps the volatility of long-dated rates. Adding it gives Hull–White,

$$dr = (\theta(t) - ar)\,dt + \sigma\,dW$$

where a is the speed at which the short rate is pulled back toward its mean, and the adjustment becomes

$$\text{Futures} - \text{Forward} = \frac{1}{\tau}\,B(T_1,T_2)\left[B(T_1,T_2)\left(1 - e^{-2aT_1}\right) + 2a\,B(0,T_1)^2\right]\frac{\sigma^2}{4a} \tag{30.13}$$

where

$$B(t_1,t_2) = \frac{1 - e^{-a(t_2-t_1)}}{a}$$

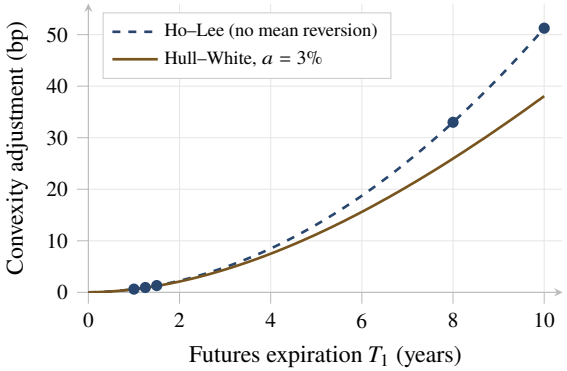
As $a \to 0$ this must reduce to (30.12), and it does: $B(T_1,T_2) \to \tau$, $B(0,T_1) \to T_1$, and $1 - e^{-2aT_1} \to 2aT_1$, so the bracket becomes $2aT_1T_2$ and the whole expression collapses to $\frac{1}{2}\sigma^2T_1T_2$. The check matters because the mean-reverting form is easy to mistype, and a wrong a is invisible at the short end and large at the long one. Equation (30.13) is the form desks most often start from; [TN30] derives both formulas.

One caution applies to both formulas. The σ in each is a normal (absolute) volatility. Substituting a lognormal volatility of, say, 20% into (30.12) produces a number four hundred times too large.

Figure 30.1 plots both versions across the strip and tabulates them at selected expirations. Two features repay attention. The first is the shape: the adjustment is almost invisible for the first two years and then accelerates, because it grows with the square of maturity. A desk can

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reasonably ignore convexity on the front contracts and cannot possibly ignore it on the back ones. The second is the gap between the curves. Mean reversion pulls long-dated rate volatility down, so the Hull–White adjustment falls progressively below Ho–Lee as maturity extends, by roughly a quarter at ten years in this calibration.



T_1 (years)	Ho–Lee (bp)	Hull–White (bp)
1	0.63	0.60
2	2.25	2.11
3	4.88	4.44
5	13.13	11.27
8	33.00	25.99
10	51.25	38.11

Figure 30.1: The futures–forward convexity adjustment against contract expiration, for the continuously compounded rate, three-month contracts and a short-rate volatility of 100 basis points per year. Markers show the values quoted in Chapter 6; the table gives both models at selected expirations.

The modern complication: averaging futures

Equation (30.12) was developed for Eurodollar futures, which settled against a *term* rate: three-month LIBOR, fixed on a single date. The contracts that replaced them work differently.

Three-month SOFR futures settle against the **compounded daily average** of the overnight rate over the reference quarter. One-month SOFR futures settle against the **arithmetic average** of the overnight rate over the calendar month. In neither case is there a single fixing date; the settlement rate accrues day by day across the contract's reference period.

This changes the risk profile in a way that matters for hedging more than for pricing:

- *The magnitude of the convexity adjustment is broadly similar.* At the leading order captured by Ho–Lee, an averaging contract and a term contract covering the same quarter carry adjustments within a basis point of each other. Equation (30.12) remains a reasonable first approximation.
- *The delta decays during the reference period.* As each day's overnight rate is observed, that portion of the settlement rate is fixed and its sensitivity disappears. A three-month SOFR futures contract's rate exposure declines roughly linearly to zero across its reference quarter. A Eurodollar futures contract's exposure, by contrast, persisted at full size until the fixing date and then vanished at once.
- *The realized volatility of the settlement rate sits between two benchmarks.* Averaging suppresses the variance of the daily fixings: Chapter 29 computes that the average carries only a third of the accrual period's variance, so it is below that of the period's last fixing but above that of a term rate fixed when the period begins, a point that matters more for options on the averaged rate than for the futures themselves.

The practical consequence is that convexity *hedging* under SOFR requires continuous rebalancing through the reference period rather than a single roll, but curve construction still requires the same order-of-magnitude correction it always did.

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Case Study 30.1 Convexity as a trade

The gap between futures and forward rates is not merely a modeling nuisance; it has long been a trade in its own right. A **convexity trade** buys futures and pays fixed on a swap covering the same period, locking in the higher futures rate on one side and the lower forward rate on the other and collecting the spread between them.

The position is not a riskless profit. The trader collects the convexity adjustment but takes on the exposure it compensates for: the trade loses when realized rate volatility exceeds the volatility implied in the adjustment. It is, in substance, a short volatility position in fixed-income form, with the risk profile that phrase implies. Convexity trades performed poorly through 2022, when realized rate volatility ran far above the levels embedded in convexity adjustments computed from the preceding decade's calm.

The general lesson recurs throughout the book: a pricing adjustment that compensates for a risk can be harvested as a return, and the return is never free.

30.4 CONVEXITY ADJUSTMENTS II: RATES PAID AT THE WRONG TIME

The second family of convexity adjustments arises when a rate is paid at a date other than the end of the period it covers. Historically this was the domain of the *in-arrears swap*; under risk-free reference rates it has changed character in an instructive way.

The classical case: rates paid in arrears

Under LIBOR conventions, a rate for $[T_1, T_2]$ was *set* at T_1 and *paid* at T_2, so the rate and its payment date were aligned. An in-arrears swap broke that alignment by paying the rate at T_1, immediately on observation.

What is now required is $E^{T_1}[R(T_1)]$ rather than $E^{T_2}[R(T_1)] = F$. The natural measure is Q^{T_2} and the required measure is Q^{T_1}, so the frame is the same as before; what is different is the density. Since $P(T_1, T_2) = 1/(1 + R(T_1)\tau)$, evaluating (30.4) at $t = T_1$ with $T = T_1$ gives

$$\Lambda = \frac{1}{P(T_1, T_2)} \cdot \frac{P(0, T_2)}{P(0, T_1)} = \frac{1 + R(T_1)\tau}{1 + F\tau}$$

which has unit mean under Q^{T_2} precisely because $E^{T_2}[R] = F$. This density is *affine* in R, and that is what makes the case exact: where Section 30.2 had to linearize a density that was not, here (30.5) can be evaluated as it stands,

$$\mathrm{Cov}^{T_2}(R, \Lambda) = \frac{\tau}{1 + F\tau}\,\mathrm{Cov}^{T_2}(R, R) = \frac{\tau\,\mathrm{Var}^{T_2}(R)}{1 + F\tau}$$

so that

$$E^{T_1}[R(T_1)] \;=\; F \;+\; \frac{\tau\,\mathrm{Var}^{T_2}(R)}{1 + F\tau} \tag{30.14}$$

exactly, with no approximation beyond the model for the variance. If R is lognormal with volatility σ, then $\mathrm{Var}^{T_2}(R) = F^2\left(e^{\sigma^2 T_1} - 1\right)$, giving

$$E^{T_1}[R(T_1)] \;=\; F + \frac{F^2\left(e^{\sigma^2 T_1} - 1\right)\tau}{1 + F\tau} \;\approx\; F + \frac{F^2\sigma^2 T_1\tau}{1 + F\tau} \tag{30.15}$$

The adjustment is positive, unlike the timing adjustment of Section 30.2, and the reason is simple: here the payment is being *accelerated*, not deferred. Receiving a rate early is worth more when rates are high, so the scenarios that get upweighted are the favorable ones.

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Example 30.2 In-arrears convexity

A swap pays the three-month rate for the period beginning in five years, but pays it at the start of that period rather than the end. The forward rate is $F = 4.00\%$ with volatility $\sigma = 25\%$; $T_1 = 5$ and $\tau = 0.25$.

The exact lognormal variance is

$$\text{Var} = F^2\left(e^{\sigma^2 T_1} - 1\right) = 0.04^2 \times \left(e^{0.25^2 \times 5} - 1\right) = 5.8694 \times 10^{-4}$$

so from (30.14) the adjustment is

$$\frac{0.25 \times 5.8694 \times 10^{-4}}{1 + 0.04 \times 0.25} = 1.453 \times 10^{-4} = 1.45 \text{ bp}$$

The rate to use is 4.0145%. The first-order approximation in (30.15), which replaces $e^{\sigma^2 T_1} - 1$ by $\sigma^2 T_1$, gives 1.24 bp—an error of 15% in the adjustment itself.

What survives under risk-free rates

A compounded RFR is determined at the end of its accrual period and paid then, so the standard product no longer breaks the alignment and the classical adjustment of Example 30.2 largely dissolves. Section 33.4 prices what is left, the timing adjustment for the few days between the end of observation and payment, and finds it a few hundredths of a basis point even at ten years. The exception is the "last reset" convention of Chapter 4, which pays a rate determined at the end of the *previous* period one period late: that is a deferral, and it carries a timing adjustment in the opposite direction to (30.14), priced with the method of Section 30.2.

30.5 CONVEXITY ADJUSTMENTS III: CMS AND THE ANNUITY MEASURE

The third convexity adjustment is the largest of the payment-timing family and, unlike Section 30.4's, was unaffected by the LIBOR transition. It concerns constant-maturity swaps.

The product and the problem

Recall from Section 28.8 that a CMS coupon pays a long *swap rate* (the ten-year rate, say) observed on each reset date T_1 and paid at the end of its accrual period, T_2. Unlike an ordinary floating leg, it pays a rate whose tenor (ten years) is not the period it is paid for (a quarter or a year).

A related and equally common structure is the **CMS spread**, paying the difference between two swap rates of different tenors (the 10-year less the 2-year, for instance). CMS spreads are the building blocks of steepener notes, which pay attractive coupons unless the yield curve flattens.

The measure problem here is structural. A swap rate S is the fixed rate that makes a swap worth zero; it equals the value of the floating leg divided by the annuity of Section 28.8 (the present value of the fixed leg per unit of fixed rate, so that one basis point on the fixed leg is worth $0.0001\,A$):

$$S(t) = \frac{\text{floating leg}(t)}{A(t)}, \qquad A(t) = \sum_{i=1}^{n} \tau_i \, P(t, t_i)$$

The annuity $A(t)$ is the price of a traded portfolio of bonds, so it is a legitimate numeraire, and S, being a traded asset divided by A, is a martingale under the associated swap measure Q^A. This is what justifies Black's formula for swaptions in Chapter 29, where the annuity appears naturally as the discounting term.

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But a CMS coupon is a *single* cash flow at a *single* date T_2. Valuing it requires $E^{T_2}[S]$, and the annuity measure supplies $E^A[S] = S_0$. The numeraires do not match, so an adjustment is required.

In the frame of Section 30.1, the natural measure is Q^A, the required measure is Q^{T_2}, and the density between them is the T_2-bond over the annuity, normalized,

$$\Lambda_t = \frac{P(t,T_2) \,/\, P(0,T_2)}{A(t) \,/\, A(0)}$$

so by (30.5) the adjustment is $\mathrm{Cov}^A(S,\Lambda)$, as in the two sections before this one. Equation (30.16) below gives the part of it for a coupon paid at T_1; moving the payment to T_2 adds the timing adjustment of Section 30.2. Evaluating that covariance needs Λ written as a function of the swap rate, and supplying it is what the next subsection's G does.

The adjustment

The correction depends on how the annuity varies with the swap rate, which is a question about the convexity of a bond price–yield relationship. Define $G(y)$ as the price, per dollar of principal, of a bond paying coupon S_0 with the swap's payment frequency and maturity, valued at flat yield y. Because the coupon equals the forward swap rate, $G(S_0) = 1$.

The trick is to adjust the bond price rather than the rate, because the bond price is the martingale. Its forward price for delivery at T_1 is a martingale under the T_1-forward measure, so $E^{T_1}[G(S(T_1))] = G(S_0)$, and expanding G to second order about S_0,

$$G(S_0) = E^{T_1}[G(S(T_1))] \approx G(S_0) + G'(S_0)\left(E^{T_1}[S] - S_0\right) + \frac{1}{2}G''(S_0)\,\mathrm{Var}^{T_1}(S)$$

the leading terms cancel, and solving for $E^{T_1}[S]$ gives

$$E^{T_1}[S(T_1)] \;\approx\; S_0 \;-\; \frac{1}{2}\,S_0^2\,\sigma^2\,T_1\,\frac{G''(S_0)}{G'(S_0)} \tag{30.16}$$

where σ is the volatility of the forward swap rate. This is for a coupon paid on the fixing date; one paid later also takes the smaller, negative timing adjustment of Example 30.1.

Since bond prices fall as yields rise, $G' < 0$; since bond prices are convex in yield, $G'' > 0$. The ratio G''/G' is therefore negative and the whole adjustment is positive. The CMS rate to use always exceeds the forward swap rate.

The intuition is the familiar one from Chapter 4's treatment of duration and convexity. The forward swap rate is set by swaps and bonds, whose prices are convex in the rate; convexity is valuable, and the market charges for it through a forward yield below the expected one. A CMS coupon pays the rate linearly and carries none of that convexity, so it is owed the rate without the charge, and (30.16) prices the difference.

Figure 30.2 shows where the adjustment comes from geometrically, and the same picture explains every convexity adjustment in this chapter. What is pinned down is the *bond price*: G is a martingale, so $E[G(S)] = G(S_0)$. Now suppose the swap rate took the values $S_0 - \delta$ and $S_0 + \delta$ with equal probability. Because G is convex, the average of G over the two outcomes lies *above* G evaluated at the average: the chord sits above the curve, which is Jensen's inequality. That would violate the martingale condition. The only way to restore it is for the distribution of S to sit at a higher level, since G is decreasing. The distance the rate must shift is precisely the convexity adjustment.

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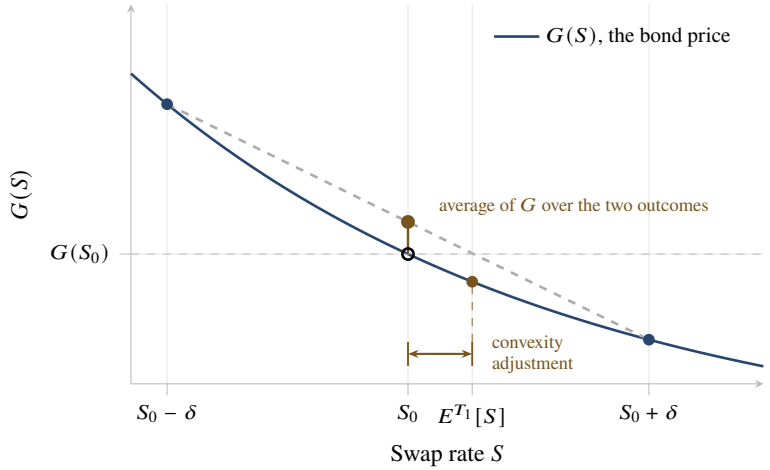


Figure 30.2: Why the expected swap rate exceeds the forward swap rate. Curvature exaggerated.

Example 30.3 CMS convexity adjustment

Consider a coupon paying the five-year swap rate observed, and paid, in five years' time. The forward five-year swap rate is $S_0 = 4.00\%$, its volatility is $\sigma = 20\%$, payments are annual, and $T_1 = 5$.

$G(y)$ is the price of a five-year annual bond with a 4% coupon. Since the coupon equals S_0, the bond prices at par when $y = 4\%$. Differentiating,

$$G'(0.04) = -4.4518, \qquad G''(0.04) = 25.0125$$

so $G''/G' = -5.6185$ and

$$E^{T_1}[S] = 0.04 - \frac{1}{2} \times 0.04^2 \times 0.20^2 \times 5 \times (-5.6185) = 0.04 + 0.000899$$

The CMS rate is **4.090%** against a forward swap rate of 4.000%—a convexity adjustment of **9.0 basis points**.

Nine basis points on a rate of four percent is a 2.2% relative correction, and it grows with the square of volatility and linearly with time to reset. On a twenty-year CMS structure the adjustment can exceed 50 basis points. Unlike the timing adjustments of Section 30.2, this is not a refinement that can be safely ignored.

CMS spreads

The structures that trade in volume are usually not single CMS coupons but *spreads* between two swap rates. Since the payoff is a difference, it is tempting to assume the adjustments on the two legs largely offset. They do not: the ratio G''/G' that drives the adjustment depends strongly on the underlying swap's maturity. A ten-year bond has far more convexity relative to its duration than a two-year bond, so the long leg carries much the larger correction.

One arithmetic simplification makes these calculations tractable by hand. Because the bond in G is constructed with coupon equal to the forward swap rate, it prices at par, and for a par bond the first derivative is exactly the negative of the annuity:

$$G'(S_0) = -A(S_0), \qquad A(S_0) = \sum_{i=1}^{n} \frac{1}{(1+S_0)^i} = \frac{1-(1+S_0)^{-n}}{S_0} \tag{30.17}$$

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The 5-year figure $G'(0.04) = -4.4518$ used in Example 30.3 is exactly the five-year annuity at 4%, and can be read off (30.17) without differentiating anything.

Example 30.4 A CMS spread coupon

A note pays the ten-year swap rate less the two-year swap rate, both observed, and paid, in five years. The forward rates and their volatilities are

$$S_0^{(10)} = 4.40\%, \quad \sigma_{10} = 18\%; \qquad S_0^{(2)} = 3.90\%, \quad \sigma_2 = 24\%$$

with annual payments and $T_1 = 5$. The forward spread is $4.40\% - 3.90\% = 50.0$ bp.

Each leg must be adjusted separately, using (30.16) with its own maturity, level, and volatility. Applying (30.17) for G', and for G'' the sum over the bond's cash flows c_i,

$$G''(S_0) = \sum_i i(i+1)\, c_i\, (1 + S_0)^{-i-2}$$

gives:

	2-year leg	10-year leg
Forward swap rate S_0	3.90%	4.40%
Volatility σ	24%	18%
Annuity $A = -G'$	1.8888	7.9518
G''	5.4189	78.3883
$-G''/G'$	2.8690	9.8580
Convexity adjustment	6.28 bp	15.46 bp
Adjusted rate	3.9628%	4.5546%

The adjusted spread is therefore

$$4.5546\% - 3.9628\% = 59.18 \text{ basis points}$$

against a raw forward spread of 50.0 basis points—a net convexity effect of **+9.2 basis points**, which is 18% of the spread being valued.

The two-year leg is assumed *more* volatile than the ten-year, as short rates typically are, which works against the effect, though nowhere near enough: the ratio $-G''/G'$ is 3.4 times larger for the ten-year leg, and that maturity mismatch dominates everything else. Had both legs referenced swaps of the same maturity, the adjustments would have largely canceled, leaving only a small net correction.

Two warnings attach to Example 30.4. First, it values the *linear* spread. Real steepener notes pay something like $\max(\text{spread}, 0)$, or a leveraged multiple of it, and an option on a difference depends on the **correlation** between the two swap rates, a quantity no convexity adjustment captures. Valuing the floored payoff requires a two-factor model or a replication argument, and the correlation involved is neither observable nor stable.

Second, the sign of the effect is systematic. Because the long leg carries the larger adjustment unless the short leg's volatility is far higher than in this example, convexity *widens* the fair spread on a steepener. A dealer who omitted the adjustment would sell every such note too cheaply, and would do so consistently rather than randomly.

Beyond the simple formula

Equation (30.16) assumes a single volatility for the swap rate, which is equivalent to assuming Black's model. Since the adjustment is fundamentally a statement about the *convexity* of the payoff, it is sensitive to the whole shape of the volatility smile, not just the at-the-money level.

Desks therefore price CMS by **static replication**: a CMS coupon decomposes into a portfolio of swaptions across all strikes, weighted by the payoff's second derivative. Write

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$H(s) = s\,\alpha_2(s)$, where $\alpha_2(s) = E^A[P(T_1,T_2)/A(T_1) \mid S(T_1) = s]$ is the annuity mapping function of Section 28.12 for payment at T_2. Then

$$V_0 = A(0)\,H(S_0) + \int_0^{S_0} H''(K)\,\mathrm{Rec}(K)\,dK + \int_{S_0}^{\infty} H''(K)\,\mathrm{Pay}(K)\,dK \tag{30.18}$$

where $\mathrm{Rec}(K)$ and $\mathrm{Pay}(K)$ are today's prices of receiver and payer swaptions struck at K: receivers below the forward swap rate, payers above, each weighted by $H''(K)$. This approach captures the whole volatility surface and is the market standard. Section 12.8 builds an arbitrary payoff from a strip of options and Section 22.3 applies the construction to variance; the CMS case is its most commercially important application in rates. Equation (30.16) remains valuable as a quick estimate and a sanity check on the replication.

Case Study 30.2 Steepeners and the flat curve

CMS spread structures were sold widely to retail and mid-market institutional investors during the 2000s, typically as notes paying a high fixed coupon for a year or two, then switching to a multiple of the 10-year-less-2-year swap spread.

Their appeal rested on the curve remaining upward-sloping. As policy rates rose (the Federal Reserve's until mid-2006, the European Central Bank's into 2008), the curve flattened and in places inverted, and the floating coupons on many of these notes fell to zero. They came back only when the rate cuts of 2007–08 steepened the curve again. Investors who had understood the note as a high-yielding bond discovered they had bought a leveraged bet on the shape of the yield curve.

The episode shows, obliquely, why this chapter's arithmetic matters. The convexity adjustment is part of the price of the curve exposure embedded in these structures, and the floor on the spread is an option whose value turns on the correlation the adjustment does not capture. An investor who could compute neither could not tell what they had been paid for the risk—and, as it turned out, few had been paid enough to justify it.

30.6 QUANTO ADJUSTMENTS

The final adjustment concerns currency rather than timing. A quanto (Section 16.7) is a derivative whose payoff depends on a variable denominated in one currency but is settled in another, at a *fixed* exchange rate specified in the contract.

A dollar-settled option on the Nikkei 225 that pays \$1 per index point gives the investor the Japanese equity return without the currency return. The dollar-denominated Nikkei 225 futures contract is the best-known listed quanto; retail notes on foreign indices and quanto CDS, which quote a sovereign's protection in a currency other than its debt's, are others. So is a euro-settled swap on dollar interest rates.

Why the FX rate matters even though it is fixed

The apparent paradox of a quanto is that the exchange rate is contractually fixed, yet exchange rate volatility affects the price. The resolution lies in hedging.

A dealer short a quanto on a foreign asset hedges by holding that asset, so the hedge's gains and losses arrive in foreign currency and are worth, at home, whatever the exchange rate is when they arrive. The quanto's payoff ignores the exchange rate. If the asset tends to rise when the foreign currency is strong, the hedge's gains are converted at a high rate and its losses at a low one, so in domestic currency the hedge systematically outruns the payoff it replicates. If the asset tends to rise when the foreign currency weakens, the hedge lags. Either way the effect recurs, a benefit or a cost according to the correlation's sign, and it must appear in the price.

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The adjustment

Adopt the convention that S is the exchange rate expressed as **units of domestic currency per unit of foreign currency** (for a US investor holding a Japanese asset: USD per JPY). The convention matters: reversing it reverses the sign of the answer. Let V be the foreign-currency price of the underlying asset, σ_V and σ_S their volatilities, and ρ the correlation between them.

The correction is the same change of numeraire as the rest of the chapter. A foreign asset's natural measure is the *foreign* risk-neutral measure Q^f, the one that goes with the foreign money market account B_f; a quanto settles in domestic currency, so the required measure is the domestic Q^d, whose numeraire is B_d. The density taking one to the other is the ratio of the two numeraires' growth, as in (30.4), but a ratio of prices needs both in one currency, and in domestic currency the foreign account is worth $S(t)\,B_f(t)$. So

$$\Lambda_t \;=\; \frac{B_d(t)\,/\,B_d(0)}{S(t)\,B_f(t)\,/\,\big(S(0)\,B_f(0)\big)} \;=\; \frac{B_d(t)\,S(0)}{S(t)\,B_f(t)}$$

which moves *against* the exchange rate. So (30.5) charges minus the covariance between the asset and S: the quanto effect takes the same form as every other adjustment here, with the sign that makes a positively correlated asset drift below r_f. [TN30] carries the derivation out in full. Under Q^d the asset's drift is therefore not its foreign risk-free rate r_f but

$$\mu_V = r_f - \rho\,\sigma_V\,\sigma_S \tag{30.19}$$

This can also be checked without the density, in two steps. Under the domestic risk-neutral measure, an investor holding foreign cash owns a traded domestic asset worth $S(t)e^{r_f t}$, whose drift must be the domestic risk-free rate r_d, so $\mu_S = r_d - r_f$, which is covered interest parity. The foreign asset's domestic value, VS, is also a traded domestic asset, and Itô's product rule gives

$$\frac{d(VS)}{VS} = (\mu_V + \mu_S + \rho\,\sigma_V\sigma_S)\,dt + \sigma_V\,dW_V + \sigma_S\,dW_S$$

whose drift must also be r_d; substituting $\mu_S = r_d - r_f$ gives (30.19). The cross-variation term $\rho\sigma_V\sigma_S$ is the entire source of the quanto effect, and it is the covariance the density charges.

The term $-\rho\sigma_V\sigma_S$ in (30.19) is the quanto drift adjustment. The measure change is the machinery, the covariance is the economics, and the hedging argument above is what the covariance means on a desk. Hence the quanto forward relates to the ordinary foreign-currency forward F_0^f by

$$\boxed{F_0^{\text{quanto}} \;=\; F_0^f\,e^{-\rho\,\sigma_V\,\sigma_S\,T}} \tag{30.20}$$

Once (30.19) is in hand, quanto derivatives are priced with entirely standard methods: use the adjusted drift, and otherwise apply the Black–Scholes–Merton model or Black's model as usual. A quanto call, for instance, is valued with Black's formula using F_0^{quanto} in place of the forward and discounting at the *domestic* rate.

Section 16.7 priced a quanto forward and call at a positive correlation, where the adjustment lowers the forward. For the dollar-settled Nikkei position above, the adjustment raises the forward: Japanese equities have tended to rise as the yen weakens ($\rho < 0$ in dollars per yen), so the dealer's hedge earns its index gains in a weakened yen, and at $\rho = -0.3$, $\sigma_V = 20\%$ and $\sigma_S = 10\%$ the one-year quanto forward sits about 0.6% above the index's ordinary forward in yen, to pay for that shortfall.

Two observations follow. First, the adjustment vanishes when $\rho = 0$: an asset uncorrelated with its own exchange rate needs no quanto correction, however volatile either may be. Second, ρ here is a correlation between an asset and an exchange rate over the life of the contract, which is notoriously unstable: the Nikkei–yen correlation has changed sign across regimes. Quanto

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positions carry a **correlation risk** that cannot be hedged with liquid instruments, and desks reserve against it instead.

30.7 A UNIFIED VIEW

Every adjustment here has been the same calculation: name the measure under which the variable's forward value is unbiased, name the measure the contract's payoff requires, write the density between them, and take the covariance. Only the density changes, as the table shows (each arrow runs from the first measure to the second).

Adjustment	Measure change	Covariance that drives it
Futures/forward convexity (30.12)	$Q^{T_2} \longrightarrow Q$ (money market)	Margin flows with the financing rate
Timing (30.8)	$Q^{T^*} \longrightarrow Q^T$	Variable with the deferral rate
In-arrears (30.14)	$Q^{T_2} \longrightarrow Q^{T_1}$	Rate with itself
CMS (30.16)	Q^A (annuity) $\longrightarrow Q^{T_2}$	Swap rate with the annuity
Quanto (30.20)	Q^f (foreign) $\longrightarrow Q^d$ (domestic)	Asset with the exchange rate

Read across the table, and the recipe is the same for any adjustment:

1. Identify the measure under which the variable's forward value is an unbiased expectation, its natural measure, determined by the numeraire that appears in its definition.
2. Identify the measure required by the contract's actual payoff date and currency.
3. Compute the covariance of the variable with whatever drives the numeraire ratio.
4. The adjustment is that covariance, to first order.

Two practical warnings close the chapter.

Adjustments are model-dependent, and the dependence is not small. Every formula here required a distributional assumption: Ho–Lee in (30.12), lognormality in (30.8), (30.15), and (30.16). Each depends on a volatility, and (30.8) and (30.20) additionally depend on a correlation. These are estimates, not observables, and the adjustment inherits their uncertainty. The 9.0 bp CMS adjustment of Example 30.3, computed at 20% volatility, becomes a 20 bp adjustment at 30% volatility, more than double. Reporting a convexity adjustment to three decimal places implies a precision that the inputs do not support.

Adjustments are systematic, not random. Unlike bid–ask spread or execution slippage, an omitted adjustment is a bias in a known direction that accumulates across every trade of its type on the book. This is why a desk that tolerates a basis point of noise will not tolerate half a basis point of bias, and why these corrections are computed and monitored rather than ignored.

SUMMARY

Forward values equal expected future values only under the measure associated with the contract's natural numeraire. When a contract's settlement mechanics, payment date, or settlement currency depart from that natural arrangement, the forward value is a biased estimate and an adjustment is required. Every such adjustment is exactly a covariance between the variable being valued and the density that changes the numeraire; it can be evaluated exactly where that density is affine in the variable, and to leading order otherwise.

Because futures contracts are settled daily, futures rates are expectations under the risk-neutral measure while forward rates are expectations under the forward measure. The resulting convexity adjustment makes futures rates exceed forward rates, by an amount growing with the square of maturity, negligible at the front of the strip and material at long maturities.

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Averaging-style RFR futures carry adjustments of similar magnitude to the term-rate contracts they replaced, but with delta that decays through the reference period.

Timing adjustments apply when a payoff is deferred or accelerated relative to its natural payment date. They are negative for deferral when the variable is positively correlated with rates, and are typically small. The transition to risk-free rates largely eliminated the classical in-arrears convexity adjustment, since a compounded RFR is determined and paid at the end of its period, but left smaller adjustments, from payment delays and from last-reset conventions that pay a stale rate one period late.

CMS convexity is the largest adjustment of the payment-timing family, arising because a swap rate is a martingale under the annuity measure while a CMS coupon is paid at a single date. It is always positive, large enough to matter in pricing, and is priced in practice by static replication with swaptions so that the full volatility smile is captured.

Quanto adjustments arise when a payoff based on a foreign variable is settled in domestic currency at a fixed exchange rate. Exchange rate volatility affects the price despite the fixed rate because it affects the dealer's hedging cost. The adjustment multiplies the foreign forward by $e^{-\rho\sigma_V\sigma_S T}$ and vanishes when the asset and exchange rate are uncorrelated.

All these adjustments depend on volatilities and correlations that are estimated rather than observed. Their importance lies less in their size than in their systematic direction: an omitted adjustment biases every affected trade the same way.

FURTHER READING

Geman, El Karoui, and Rochet (1995) is the source of the change-of-numeraire results this chapter runs on. Brigo and Mercurio's *Interest Rate Models—Theory and Practice* treats convexity adjustments and CMS pricing in full in its Chapter 13. Kirikos and Novak (1997) derive the Hull–White futures convexity adjustment, and Hagan's "Convexity Conundrums" (2003) is the standard practitioner treatment of static replication for CMS.

PRACTICE QUESTIONS

30.1 Explain why a forward rate is a martingale under the T_2-forward measure but not under the risk-neutral measure. Which measure is relevant for valuing a payoff made at T_2?

30.2 Without performing any calculation, state the sign of the required adjustment in each case, and justify your answer in one sentence:

(a) a variable uncorrelated with interest rates, paid three months after its natural date;

(b) a ten-year swap rate paid as a single cash flow in five years;

(c) a quanto forward on a foreign equity index, where the index and the foreign currency are positively correlated.

30.3 A three-month interest rate futures contract has $T_1 = 3$ years and $T_2 = 3.25$ years. The short-rate volatility is 90 basis points per year. The futures rate is 3.85%. Calculate the forward rate using the Ho–Lee adjustment.

30.4 Repeat Question 30.3 with $T_1 = 8$ and $T_2 = 8.25$. By what factor does the adjustment increase, and does that match what equation (30.12) predicts?

30.5 Explain in words why a trader who is long the rate through a futures position benefits from daily settlement, and why this requires the futures rate to exceed the forward rate.

30.6 A payoff equal to the two-year swap rate is observed in three years and paid six months later. The two-year swap rate's expected value for payment at year 3 (its forward plus the CMS adjustment) is 3.50% with 22% volatility; the forward rate for the deferral period is 3.20% with 25% volatility; the correlation is 0.8. Calculate the timing adjustment in basis

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points.

30.7 A rate for a six-month period beginning in four years is paid at the start of the period rather than the end. The forward rate is 3.75% with volatility 28%. Calculate the in-arrears convexity adjustment using both the exact form (30.14) and the approximation in (30.15), and comment on the difference.

30.8 Explain why the transition from LIBOR to compounded risk-free rates eliminated most of the in-arrears convexity adjustment. Identify two situations in which a convexity adjustment is still required under RFR conventions.

30.9 A quanto forward on a foreign stock index is settled in domestic currency. The index has 25% volatility, the exchange rate (domestic per foreign) has 12% volatility, and the correlation is 0.4. The one-year foreign-currency forward level is 5,000. Calculate the quanto forward level.

30.10 Continuing Question 30.9, the domestic interest rate is 3%. Value a one-year quanto call on the index, struck at the quanto forward level and paying one unit of domestic currency per index point, and compare it with the same call if the correlation were zero. Why does the correlation matter when the contract's exchange rate is fixed?

30.11 Without using algebra, explain why the convexity adjustments on the two legs of a CMS spread do not cancel. Under what condition would they cancel?

FURTHER QUESTIONS

30.12 A CMS coupon pays the ten-year swap rate observed in three years. The forward ten-year swap rate is 4.20%, its volatility is 18%, and payments are annual. Compute G' and G'' for a ten-year annual bond with a 4.20% coupon at a flat yield of 4.20%, and hence find the convexity adjustment. Compare the size of your answer with Example 30.3 and explain the difference.

30.13 Using the result of Question 30.12, recompute the adjustment at volatilities of 12% and 24%. Comment on what your answers imply about the precision with which convexity adjustments can be quoted.

30.14 A dealer sells a quanto call on a foreign index. Describe the dealer's hedge, explain how the hedge must be rebalanced as the index moves, and use this to give an economic derivation of the sign of the quanto adjustment in equation (30.20). Why can this position not be fully hedged in practice?

30.15 A structured note pays $3 \times$ (10-year swap rate $-$ 2-year swap rate) annually for ten years, subject to a floor of zero. Identify every adjustment from this chapter that its valuation requires, and explain which of them dominates.

30.16 Show that a rate futures contract's convexity adjustment can be interpreted as the value of an option-like exposure to realized rate volatility. Using this interpretation, explain why a convexity trade that appeared profitable on 2019 volatility estimates would have lost money through 2022.

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